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Cummins Inc. - Branded Bosch® Starters and Alternators

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This Service Bulletin covers repair kits, standard repair times, service tools and literature to support Cummins Inc. - Branded Bosch® Alternators and Starter Motors, for the B,C,L,M & N Series engines.

Note : There are no prescribed service intervals for the Cummins Inc. - Branded Bosch® Alternators and Starter Motors.

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Section 1. Operating Guidelines

When performing starter or alternator service or repairs, general safety precautions **must** be followed to provide personal safety, as outlined in Cummins Inc. Service Manuals.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Acid is extremely dangerous and can damage the machinery and can also cause serious burns. Always provide a tank of strong soda water as a neutralizing agent when servicing the batteries. Wear goggles and protective clothing to reduce the possibility of serious personal injury.

⚠ CAUTION ⚠

To prevent damage to the starter, do not engage the starting motor for more than 30 seconds at a time.

Wait 2 minutes between each attempt to start (electrical starting motors **only**).

If the engine does **not** start after three attempts, check the fuel supply system.

Absence of blue or white exhaust smoke during cranking indicates that fuel is **not** being delivered. The starter **must not** be used to prime the fuel system. Reference the appropriate base engine Troubleshooting and Repair Manual for correct priming procedures.

The starter is an intermittent duty electric motor that is capable of high outputs for short periods of time. Excessive cranking of the starting motor will lead to an unwarrantable failure.

Section 2. Cummins to Bosch Part Number Cross Reference

This section tabulates the Cummins, ReCon® and Bosch® part number reference for the branded starters and alternators.

Table 1, Cummins Inc., ReCon® and Bosch® Part Number Reference

Cummins Part Number	Bosch® Branded Part Number	Component	Voltage and Power	Description
3255439	0 120 469 019	Alternator	28VDC/55A	N1 L Design Standard
3276269	0 120 469 562	Alternator	28VDC/55A	N1 Standard
3282554	0 120 484 011	Alternator	14VDC/95A	K1 L Design
3283780	0 120 469 796	Alternator	24VDC/55A	N1 Marine Standard
3283781	0 120 469 040	Alternator	14VDC/90A	N1 Marine Standard
3283809	0 120 689 543	Alternator	28VDC/120A	T1 Long Life Bearing
3283810	0 120 689 540	Alternator	24VDC/120A	T1 Long Life Bearing
3283811	0 120 468 137	Alternator	28VDC/80A	N1 Standard

3920678	0 120 488 206	Alternator	28VDC/45A	K1 L Design
3920679	0 120 488 205	Alternator	14VDC/65A	K1 L Design
3923624	0 120 468 054	Alternator	28VDC/65A	N1 Long Life Bearing (SAE)
3283329	0 001 362 319	Starter	12VDC/2.7kW	JF
3283330	B 001 816 285	Starter	12VDC/3.0kW	JF Long Life Clutch
3283721	0 001 388 067	Starter	24VDC/4.0kW	JF Isolated Return/Long Life Clutch
3283722	B 001 816 421	Starter	12VDC/3.0kW	JF Isolated Return/Long Life Clutch
3283812	0 001 369 019	Starter	12VDC/3.1kW	JF

Section 3. Repair Kits

Note : Individual components within the kits will not be made available.

The following Repair Kits are used for the repair of the Cummins Inc. - Branded Bosch® Alternators and Starter Motors.

Table 2, Alternator Repair Kits

Cummins Part Number	Parts Set (800) Armature	Carbon-Brush Set (17/801)	Parts Set (804) Terminal	Collector Ring (807)	Fan (807)	Air-Intake Cover (713)	Grommet (722)	Regulator
3255439	3870961	3870965	3870969	3870973	N/A	N/A	N/A	3870982
3276269	3870961	3870965	3870969	3870973	N/A	N/A	N/A	3870982
3282554	3870960	3870965	3870967	3870974	N/A	N/A	N/A	3870979
3920678	3870960	3870966	3870967	3870973	N/A	N/A	N/A	3870980
3920679	3870960	3870965	3870967	3870972	N/A	N/A	N/A	3870979
3923624	N/A	3870965	3870968	3870973	N/A	N/A	N/A	3870981
3283780	3870961	3870971	3870969	3870973	3870976	N/A	N/A	3870984

328378 1	387096 1	387096 5	387096 9	387097 3	387097 6	N/A	N/A	387098 3
328380 9	387096 2	N/A	387097 0	387097 5	N/A	387097 7	387097 8	387098 5
328381 0	387096 3	N/A	387097 0	387097 5	N/A	387097 7	387097 8	387098 5
328381 1	387096 4	387096 5	387096 9	387097 3	N/A	N/A	N/A	387098 2
	Multiple Parts	Multiple Parts	Multiple Parts	Single Part	Single Part	Single Part	Single Part	Single Part

Table 3, Starter Motor Repair Kits

Cummins Part Number	Carbon-Brush Set (800)	Parts Set (802)	Fork Lever (803)	Solenoid (804)	Drive End Shield (805)	Flat-Head Screw (807)	Parts Set (810)
3283329	3870986	3870989	3870990	3870991	3870994	3870996	3870998
3283330	3870997	3870989	3870990	3870991	3870994	3870996	3871000
3283721	3870988	3870989	3870990	3870993	3870994	3870999	N/A
3283772	3870987	3870989	3870990	3870992	3870994	3870996	N/A
3283812	3870987	3870989	3870990	N/A	3870995	3870997	3870999
	Multiple Parts	Multiple Parts	Single Part	Single Part	Single Part	Single Part	Multiple Parts

Section 4. Warranty Information

A. Cummins Branded Bosch Starters and Alternator Service and Warranty

Note : This is the sole and exclusive warranty applicable to these products. There are no other warranties, expressed or implied, including no warranties of fitness for a particular purpose and merchantability. Under no circumstances will Cummins Inc. be liable for incidental or consequential damage.

All Cummins Inc. Plants phased in the introduction of Branded Bosch® starters and alternators fitted to engines from September 1st, 1997. These starters and alternators will have new dataplates which state that this product is built for Cummins Inc. by the accessory supplier.

These products will be covered under New Engine Warranty on all applications except for the U.S./Canada Recreational Vehicle (RV), School Bus, Fire and Crash Truck where the coverage is limited to 2 years. The appropriate Legal Description has been revised to reflect this change of coverage. This warranty applies to the Cummins Inc. - Branded Starters and Alternators **only**. If a Cummins Inc. - Branded Starter or Alternator fails during the warranty period due to a defect in material or workmanship, Cummins Inc., at its option, will replace or repair the starter or alternator without charge to the owner.

In order to receive warranty reimbursement for Cummins Inc. - Branded Starters and Alternators, the Cummins Inc. branded dataplates **must** be confirmed.

The use of Recon® is the standard method for replacing the failed components. The warranty tag in the Recon® box is to be completed, attached to the failed core and returned to Cummins UKDRC for the UK and MPDC for Europe, Middle East and Africa. The warranty tag has to match with the submitted warranty claim. Repairs, instead of Recon® exchange, can be carried out if the cost is less, or if usage of Recon® is **not** feasible. For this, Cummins Inc. issued Rotary Electrics repair kits have to be used in connection with proper tooling and trained personnel. The Standard Repair Times for starters and alternators have been updated to reflect the use of these products.

B. Warranty Guidelines

The warranty of the Cummins Inc. - Branded Starters and Alternators will be administered according to the guidelines in the Cummins Warranty Administration Manual, Bulletin 4021290 (Cummins Form 6299).

Starters and alternator failures that are **not** considered warrantable are:

- 1. Customer abuse, including but not limited to:
 - Handling damage
 - Stripped or overtightened threads
 - Contamination failures
 - Cranking with low system voltage
 - Excessive cranking of the starter
 - Alternator failure due to low or defective batteries.
- 2. Corrosion
- 3. Customer or OEM installation and wiring issues:
 - Magnetic switches located with a wrong orientation or a stuck switch
 - Improperly sized cables
 - Poor ground connections
 - Loose cables or connections
 - Loose customer installed mounting bolts
 - Loose customer installed alternator pulley
 - Milling of pinions.
- 4. Customer misapplication
- 5. No Trouble Found starters and alternators.

Standard Repair Times

Table 4, Standard Repair Times - Starters		
Procedure Number	Description	Hours

13-606	General overhaul	1.6
13-608	Disassembly, cleaning, examination	0.7
13-609	Repair and assembly	0.7
13-607	Testing	0.2
Table 5, Standard Repair Times - Alternators		
Procedure Number	Description	Hours
13-128	General overhaul	1.5
13-130	Disassembly, cleaning, examination	0.7
13-131	Repair and assembly	0.5
13-129	Testing	0.2

Section 5. Service Literature

Cummins Inc. Service Bulletin 3666625 has been released to support the branded Prestolite/Bosch® starters and alternators. Reference the following (Table 6) for Bosch® test and repair instructions:

Table 6, Prestolite/Bosch® Starters and Alternators	
Branded Bosch Starters	12VDC/24VDC JF DESIGN
Branded Bosch Alternators	24VDC/28VDC N1 DESIGN
Branded Bosch Alternators	14VDC N1 DESIGN
Branded Bosch Alternators	14VDC K1 DESIGN
Branded Bosch Alternators	28VDC K1 DESIGN
Branded Bosch Alternators	24VDC/28VDC T1 DESIGN

This Service Bulletin 3666644, is intended to be used with the following:

Service Information Binder - Branded Rotary Electrics (Prestolite/Bosch®), Bulletin 3666625.
Warranty Adjudication Posters (alternators), Bulletin 3666637.
Warranty Adjudication Posters (starters), Bulletin 3666638.

Section 6. Service Tools

This section outlines the tools required to troubleshoot and diagnose starter and alternator failures. It is divided into two categories. The first category is for tools that are required for on engine diagnostics. The second category is for over the counter exchanges and rebuilding.

Table 7, Category 1 - Required On Engine Diagnostics	
Description	Service Tool Part Number

Multimeter	3377161
Clamp on Current Probe	3823574
Carbon Pile Load Tester	3885847
Digital Optical Tachometer	3377462
Table 8, Category 2 - Over the Counter Exchange or Rebuild	
Description	Service Tool Part Number
Bench Tester, 3 phase 400 VAC, 50 Hz	3825047
Bench Tester, 3 phase 440 VAC, 60 Hz	3825048

Section 7. Service Training

Service Training relevant to testing and troubleshooting the branded Bosch® starters and alternators is available.

Details can be obtained from:

**Service Training Centre.,
Customer Support Group
Cummins Engine Company Ltd.
Royal Oak Way South,
Daventry, Northants, NN11 5NU.
ENGLAND
Telephone: (44) 01327 886000
Facsimile: (44) 01327 886118**

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